

# Theory of Computation

BSc Computer Science - Durham University

Matthew Emerson

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# 1 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

## 1.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 1.

| Title       | Opening  | Due | Feedback   | Professor                    | Weight |
|-------------|----------|-----|------------|------------------------------|--------|
| Benchtest 1 | Week 11  | n/a | 09/02/2027 | Dr Dantchev and Dr Mertziros | 17%    |
| Benchtest 2 | Week 20  | n/a | 07/05/2027 | Dr Dantchev and Dr Mertziros | 17%    |
| Final Exam  | May Exam | n/a | n/a        | Dr Barnaby Martin            | 66%    |

Table 1: Assessments for Theory of Computation.

## 1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

| Day       | Time          | Class Type    | Location | Lecturer          |
|-----------|---------------|---------------|----------|-------------------|
| Tuesday   | 10:00 - 11:00 | Lecture       | CLC202   | Dr Barnaby Martin |
| Wednesday | 09:00 - 11:00 | Problem Class | TLC116   | n/a               |
| Thursday  | 14:00 - 15:00 | Lecture       | CLC202   | Dr Sukanya Pandey |

Table 2: Michaelmas Term Timetable for Theory of Computation.

# 2 Models of Computation

This section is taught by Dr Barnaby Martin.

## 2.1 Finite-State Automata and Regular Languages

## 2.2 Büchi Automata and Omega-Regular Languages

## 2.3 Linear-Time Logic and Translating Into Büchi Automata

# 3 Algorithms and Complexity

## 3.1 Asymptotic Complexity and Notation

## 3.2 Greedy Algorithms

## 3.3 Divide and Conquer Algorithms

## 3.4 Dynamic Programming Algorithms

## 3.5 Machinings in Graphs

## Part I

# Appendix

## 4 Bibliography

### 4.1 Module Professors

**Module Lead:** Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

**Lecturer:** Doctor Sukanya Pandey

- Focus: Not specified.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

**Lecturer:** Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

### 4.2 References

### 4.3 List of Figures

### 4.4 Recommended Reading

Cormen,T.H., Leiserson,C.E, Rivest,R.L, and Stein,C. (2009) *Introduction to Algorithms - Third Edition*. MIT Press.